



1.11

Equivalent Representations of Polynomial and Rational Expressions

Student Notes and Guided Practice

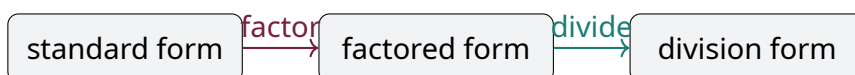
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Rewrite polynomial and rational expressions into strategically useful forms.
- Use factoring, expansion, and polynomial division to reveal zeros and asymptotes.
- Track restrictions when simplifying rational expressions.

Essential Question

Which equivalent representation reveals the feature needed for a particular question?



degree and coefficients zeros, multiplicities, cancellations, holes behavior and asymptotes

Concept Framework

Key Idea

Standard form highlights degree and leading coefficient.

Key Idea

Factored form highlights zeros, multiplicities, cancellations, and sign changes.

Key Idea

Division form highlights polynomial asymptotes and remainders.

Key Idea

Equivalence is algebraic; equality as functions also requires matching domains.

Formula Map**Formula and Structure**

$$P(x) = a_n x^n + \cdots + a_0$$

Formula and Structure

$$P(x) = a \prod (x - r_i)^{m_i}$$

Formula and Structure

$$\frac{P}{Q} = S + \frac{A}{Q}$$

Worked Examples**Worked Example 1**

Problem. Rewrite $x^3 - 4x^2 - x + 4$ in factored form.

Solution. Group: $x^2(x - 4) - 1(x - 4) = (x^2 - 1)(x - 4) = (x - 1)(x + 1)(x - 4)$.

Worked Example 2

Problem. Divide $2x^3 + 3x^2 - 5$ by $x - 1$.

Solution. The quotient is $2x^2 + 5x + 5$ with remainder 0, so the expression factors as $(x - 1)(2x^2 + 5x + 5)$.

Worked Example 3

Problem. Why is $\frac{x^2-1}{x-1}$ not the same function as $x + 1$?

Solution. The first function excludes $x = 1$; the second includes it.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Rewrite $x^3 - 9x$ into a form that reveals zeros or restrictions.

2. Rewrite $x^4 - 16$ into a form that reveals zeros or restrictions.

3. Rewrite $2x^3 + x^2 - 8x - 4$ into a form that reveals zeros or restrictions.

4. Rewrite $\frac{x^2+5x+6}{x+2}$ into a form that reveals zeros or restrictions.

5. Rewrite $\frac{x^3+1}{x+1}$ into a form that reveals zeros or restrictions.

6. Divide $x^4 - 3x^2 + 2x - 5$ by $x - 2$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.